Subarray Sum Equals K (medium)

Problem Statement

Given an array `nums` containing `n` integers and integer `k`, return the total number of `subarrays` having `sum` equal to `k`.

A `subarray` is defined as a contiguous `non-empty` sequence of the array elements.

Examples

Example 1:

- **Input**: `nums = [1, 2, 3], k = 3`
- **Expected Output**: `2`
- **Justification**: There are two subarrays that sum to 3: `[1, 2]` and `[3]`.

Example 2:

- **Input**: `nums = [10, 2, -2, -20, 10], k = -10`
- **Expected Output**: `3`
- **Justification**: Three subarrays sum up to -10: `[10, 2, -2, -20]`, `[2, -2, -20, 10]`, and `[-20, 10]`.

Example 3:

- **Input**: `nums = [5, 1, 2, -3, 4, -2], k = 3`
- **Expected Output**: `2`
- **Justification**: There are two subarrays that sum to 3: `[2, -3, 4]`, and `[1, 2]`.

Try it yourself

Try solving this question here:

Java

```java
import java.util.HashMap;
import java.util.Map;

public class Solution {
    public int subarraySum(int[] nums, int k) {
        // ToDo: Write Your Code Here.
        return 0;  // Return the total count of subarrays
    }
}
```

Test Case | Run Code Result | Stdout

```
[1,1,1]
2
```

Use Example Testcases | Run | Submit